PRD: Crypto Signal & Learning Platform (v2, merged)

Status	Draft v2: source PRD + owner decisions
Date	2026-11-06
Owner	Syubban
Source	`resources/PRD Crypto Signal & Learning Platform.md` (authoritative for the Signal Checker and its logic)
Reference	`blockchain-made-easy` (lesson-module pattern only, not its look)

This document replaces my earlier draft. Anything not changed here is defined in the source PRD.

1. Changes versus the source PRD

#	Change	Reason
C1	Learn lessons use **interactive charts**, not only static SVG.	Owner wants to learn by manipulating the graph.
C2	**Lesson progress** is stored in `localStorage` (key `cme:v1`).	Owner decision.
C3	The Signal Checker **stays stateless**: inputs, results and plans are never stored. A refresh clears them.	Preserves source acceptance criterion 6.
C4	Visual design: **calm, minimal, light theme**. Not the look of `blockchain-made-easy`.	Owner decision and source PRD.
C5	Lesson module pattern borrowed from `blockchain-made-easy`: ordered chapters, glossary hover terms, completion mark.	Owner's original request.

All other scope, non-goals (no live data, no accounts, no execution, spot only, no prediction) and the four signal states are unchanged.

2. Product

A static English web app for casual spot swing traders (4H/1D) who cannot read exchange charts such as Gate.com. Two pillars:

1. **Learn**: short chapters, each with an interactive chart that shows what a chart element means.
2. **Signal Checker**: type in numbers from the exchange chart; get Buy Zone / Wait / Take Profit / Plan Broken, the 3 Lights, and a calm plan (3-tier entry ladder, cash reserve, stop-loss, take-profit).

Promise: the user always knows where the floor is, how much cash is left, and what the plan is before placing an order.

3. Learn (changed: interactive)

Chapters follow the source's 8 topics, each with one interactive component. Each chapter ends with a link to the Signal Checker and a "Mark as done" control.

#	Chapter	Interactive element
1	How a chart is built	Three-stage cause-and-effect: toggle order book, then candles + volume, then indicators on a sample chart; hotspots on a generic exchange-style screen.
2	Candlestick anatomy	Drag open/high/low/close handles; body and wicks redraw; preset shapes (long lower wick at support, etc.).
3	Indicators in plain language	Toggle MA (or EMA), BOLL, RSI, VOL, MACD on a sample chart; period sliders.
4	Support, resistance, liquidity zones	Click to place floors and ceilings; see bounces and breaks.
5	The 3-Light checklist	Sliders for price/MA30, RSI, Bid %; the three lights and resulting state update live; worked example.
6	Position sizing & split entry	Capital slider; 3 tiers plus reserve drawn on a chart with the stop level.
7	Trading mindset	Same chart replayed as "market breathing" (5% to 7% dip above the stop) vs "broken trend" (below the stop).
8	Glossary	Every abbreviation with an everyday analogy; terms also appear as hover definitions inside chapters.

Rules:

- Sample charts use synthetic, deterministic candle data labelled as an example.
- Source-document price levels (BTC/ETH/SOL, 5 to 6 November 2026) appear only as dated worked examples.
- Every chart has a text or table alternative; every drag has a keyboard equivalent.

4. Signal Checker (unchanged from source, plus implementation defaults)

Inputs, validation, signal states, ladder ratios (about 13 / 21 / 30 / 36 percent), stop-loss, result screen order and disclaimer are exactly as in the source PRD sections FR-1 to FR-5 and FR-7.

Open questions in the source PRD get these **proposed defaults** (owner to confirm; each is one constant in code):

Source open question	Proposed default
Sell-side thresholds	Price above MA30 or at/above BOLL upper; RSI > 70; Bid % < 30
Lights needed for Sell	Take Profit if price reached the take-profit target, or if at least 2 of 3 mirrored lights are on
Stop-loss buffer	2% below the lowest support entered
Buy RSI light	RSI ≤ 36
Take-profit default	Nearest resistance, editable; show the % gain from Entry 1
Product name	Keep "Crypto Made Easy" as working name

Signal logic lives in pure TypeScript functions with unit tests for all six source acceptance cases.

5. Storage & privacy

- `localStorage` holds only `{ schemaVersion: 1, completed: string[] }` for lesson progress.
- Nothing the user enters in the Signal Checker is stored or transmitted.
- Corrupted storage falls back to an empty state without crashing.
- A "Reset progress" action with confirmation.
- No cookies, no analytics, no network calls with user data.

6. Design (calm, minimal, light)

- Light theme only in v1, generous spacing, one accent colour reserved for signals.
- Signal state is never colour alone: icon plus text on every badge.
- Charts are the visual hero; avoid card grids, gradients and decoration. Candle direction also encoded by shape/label, not red/green alone.
- Visual direction produced by the project's design skills (`frontend-design` , `anti-ui-slop` , `design-taste-frontend` app-screen rules) into a short design contract before UI work.
- Mobile-first; the whole checker usable on a phone.

7. Technical approach

Area	Decision
Base	Existing `ts-astro-starter` (Astro 7, strict TS, plain CSS tokens, Vitest, Playwright, ESLint, CI), demo content removed
Interactivity	React islands (`client:visible` / `client:load` for the checker)
Charts	`lightweight-charts` for market-style charts; custom SVG for candle editor and diagrams
Content	MDX lessons in a content collection (`track` , `chapter` , `terms`)
State	Local component state; progress via a tiny `localStorage` module
Styling	Plain CSS custom properties (no Tailwind/Starlight)
Hosting	Any static host (Cloudflare Pages suitable)

```
src/
  lib/signal/        lights.ts state.ts ladder.ts stop.ts validate.ts
  lib/indicators/    sma ema rsi macd bollinger
  lib/candles/       generator.ts aggregate.ts
  lib/progress.ts
  components/checker/  CheckerForm ResultScreen LightsPanel PlanCard
  components/charts/   MarketChart CandleEditor
  components/lessons/  one island per chapter
  content/lessons/     01-how-a-chart-is-built.mdx ...
  pages/               index, checker, learn/index, learn/[slug], glossary, about
tests/unit/            signal, indicators, progress
```

8. Non-functional requirements

Performance: fast static load, minimal JS on lesson pages without an island, charts lazy-loaded. Accessibility: AA contrast, keyboard-operable form and chart controls, alt text for diagrams. Browsers: latest two of Chrome, Safari, Firefox, iOS Safari. Language: English.

9. Acceptance criteria

Source criteria 1 to 6 unchanged (SOL Buy Zone, ETH Wait, 233.77 ladder, Plan Broken, RSI 120 validation, refresh clears inputs), plus:

1. All 8 chapters render; chapters 1 to 7 each have a working interactive chart.
2. Marking a chapter done persists across reloads; reset clears it.
3. The Checker stores nothing: after refresh, all inputs are empty.
4. Candle editor enforces high ≥ max(open, close) and low ≤ min(open, close).
5. Every result shows stop level, reserve, "what to do" and "what not to do", plus the disclaimer.
6. Every signal badge shows icon and text, not colour alone.

10. Risks

Risk	Mitigation
False confidence from "Buy Zone"	Disclaimer, mandatory stop, checklist wording
Buy-on-weakness fails in a downtrend	Plan Broken state and stop
Synthetic charts look unrealistic	Tune generator (volatility clustering, wicks)
Interactive charts hard on touch	Pointer events, large hit areas, keyboard fallback
Scope creep from interactivity	One interactive component per chapter, shared chart components

11. Remaining questions

1. Confirm the proposed defaults in section 4.
2. Quizzes per chapter in a later version?
3. Public launch and domain?